

Open Source Platforms You Can Use for AR and VR



Augmented reality and virtual reality are being used in various domains and have many useful applications today, apart from gaming. A few useful open source platforms that help to implement these technologies are listed in this article.

The global market for augmented reality (AR) and virtual reality (VR) is increasing at an exponential rate. Consumers and enterprises are benefiting, as they can access and experience a remote environment in real-time.

Figure 1 depicts Statista.com's research analytics with respect to the AR and VR market, both from historical as well as future perspectives.

Virtual and augmented reality complement each other, as they enhance people's perception of their surroundings by incorporating digital elements that look like natural features. Virtual Fixtures, a system created by the US Air Force's Armstrong Laboratory in 1992, was the first fully functional AR system that enabled an immersive mixed reality experience for users.

The entertainment and gaming industries were the first to offer commercial augmented reality experiences. Many commercial businesses have adopted AR applications for education, healthcare and entertainment since then. With AR techniques, content can even be accessed by scanning or viewing a picture with a mobile device.

AR provides users with new levels of sensory stimulation. Its usage (e.g., integrating computer vision, putting AR cameras into smartphone applications, and object identification) can help you digitally modify and interact with the surroundings. An example is superimposing additional measured information like electromagnetic radio waves, in precise alignment with where they are in space, to the existing

reality. There is a huge potential for AR to be used in the collection and dissemination of tacit knowledge. Real-time and semantic settings that contain environmental factors are frequent when using augmentation techniques. As an example, a live video broadcast of a sporting game can provide both immersive and perceptual information along with additional data like scores. Technologies that use both AR and heads-up display (HUD) technology can be combined in this manner.

AR vs VR

Virtual information permeates the user's experience of reality in VR. In AR, the user's view of reality is enriched by including computer-generated information inside the data gathered from the actual world.